



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

CIVICS GROUP

C	G			/	2	0
---	---	--	--	---	---	---

H2 BIOLOGY

Paper 3

Long Structured Question and Free-response Questions

9744/03

14 September 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/ 25
2	/ 13
3	/ 12
Section B	/ 25
Total	/ 75

Section A

Answer **all** the questions in this section.

- 1 The arginine (*arg*) operon in *E. coli* is a repressible system with three genes, *arg B*, *arg C* and *arg H*, that code for enzymes responsible for the synthesis of the amino acid arginine. The *arg* operon is shown in Fig. 1.1.

Another gene upstream of the operon (not shown in Fig. 1.1) codes for a repressor protein R. It binds to the operator to regulate the operon. Arginine binds to repressor R and activates it.

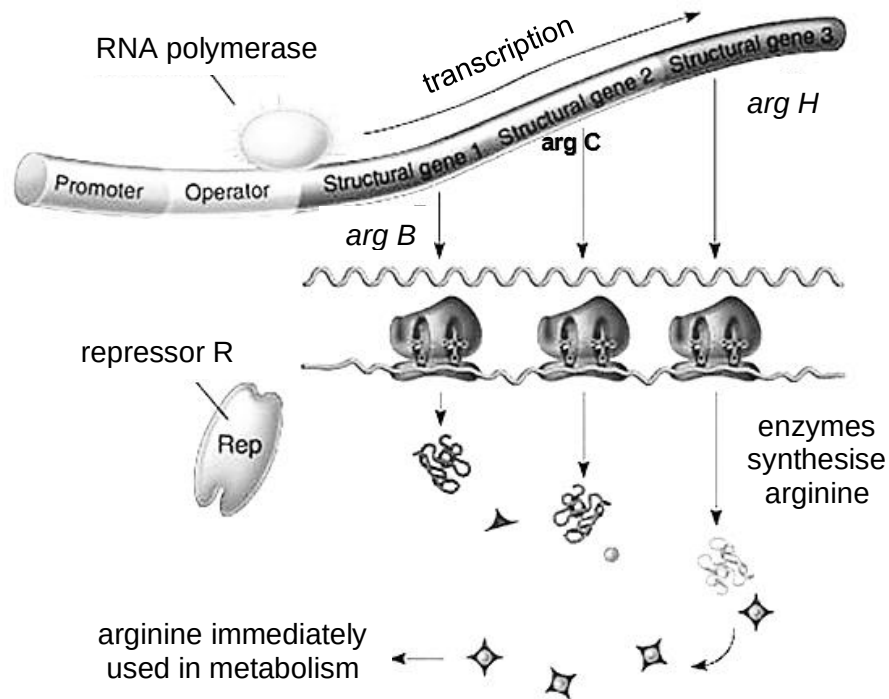


Fig. 1.1

- (a) (i) Explain why the *arg* operon is a repressible system.

.....

.....

.....

.....

..... [2]

- (ii) Describe one similarity and one difference between the repressible system shown by *arg* operon and the end product inhibition of enzyme activity in bacteria.

similarity

.....

.....

difference

.....

..... [2]

Mutations were introduced into *E. coli* strains **E1**, **E2** and **E3**. The activity of *arg B* gene, in the form of presence or absence of enzyme activity, was measured in different *E. coli* strains.

The results are shown in Table 1.1.

Table 1.1

<i>E. coli</i> strain	genotype	presence (✓) or absence (✗) of enzyme activity	
		arginine present	arginine absent
E1	R ⁺ O ⁺ B ⁻	✗	✗
E2	R ⁻ O ⁺ B ⁺	✓	✓
E3	R ⁺ O ⁻ B ⁺	✗	✗

Key: R⁺ – normal repressor gene *R*
R⁻ – mutated repressor gene *R*

O⁺ – normal operator
O⁻ – mutated operator

B⁺ – normal *arg B* gene
B⁻ – mutated *arg B* gene

- (b) With reference to Table 1.1, explain the abnormal *arg B* gene activity in strains **E2** and **E3**.

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Bacteria may be exposed to different types of stress, and hence had developed appropriate responses in order to survive. A variety of heat shock proteins (HSP) that are expressed in response to changes in temperature have been identified from *E. coli*.

To study the expression levels of the different HSP genes, bacterial cells were subjected to heat stress of 45°C. Proteins were then extracted from *E. coli* and separated by two-dimensional (2D) gel electrophoresis. This technique is carried out in a sequence shown in Fig. 1.2.

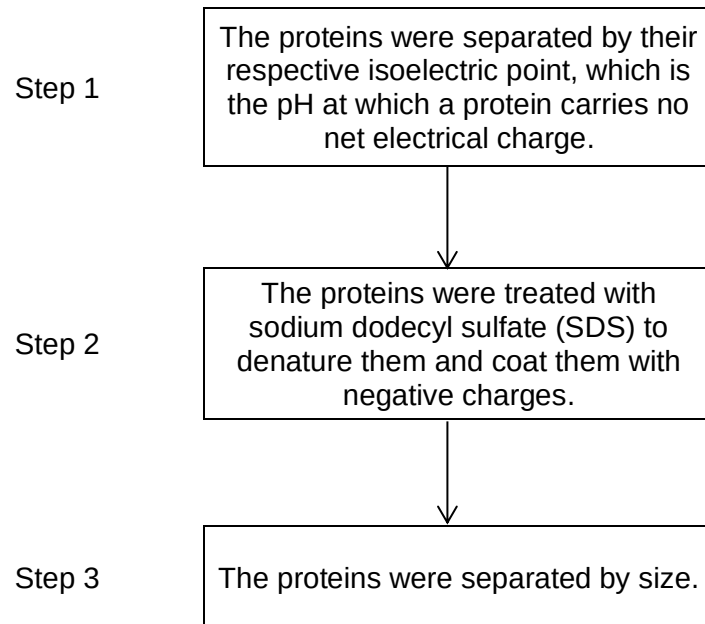


Fig. 1.2

- (i) Suggest why SDS treatment is required in step 2.

.....

 [1]

The result of the 2D gel electrophoresis is shown in Fig.1.3. The size and intensity of each protein spot on the gel corresponds to its amount.

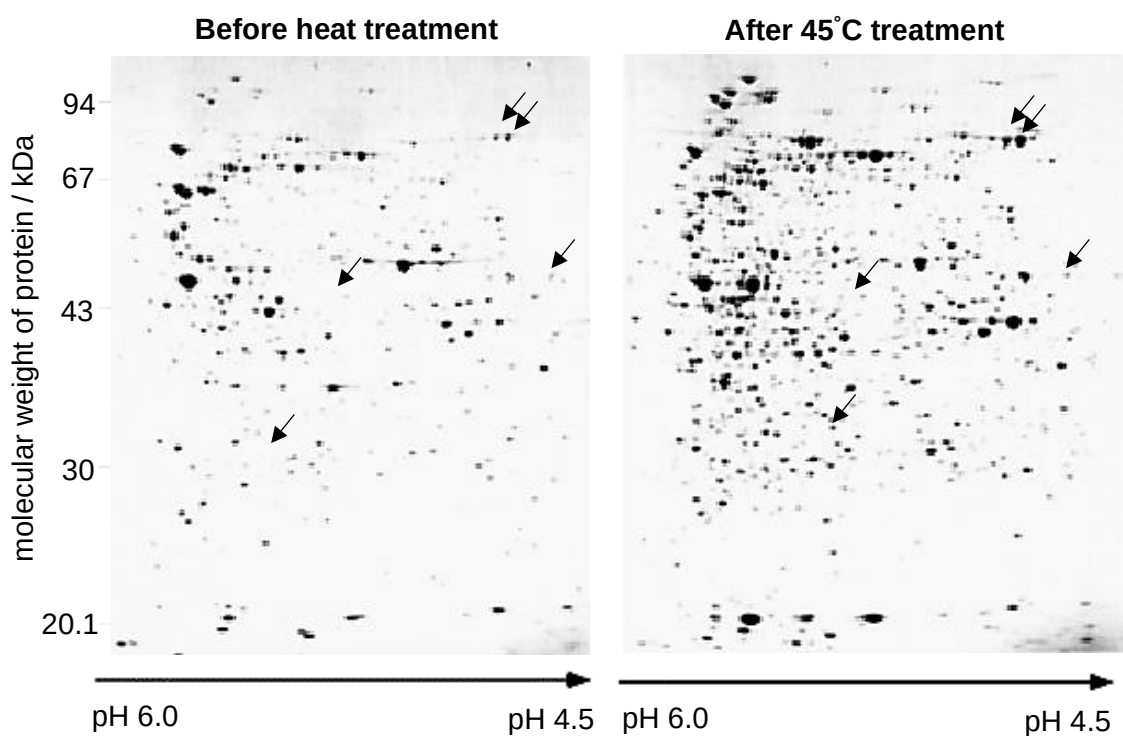


Fig. 1.3

The experiment was carried out six times and the expression levels of five types of HSP (arrowed) before and after heat treatment is shown in Fig 1.4.

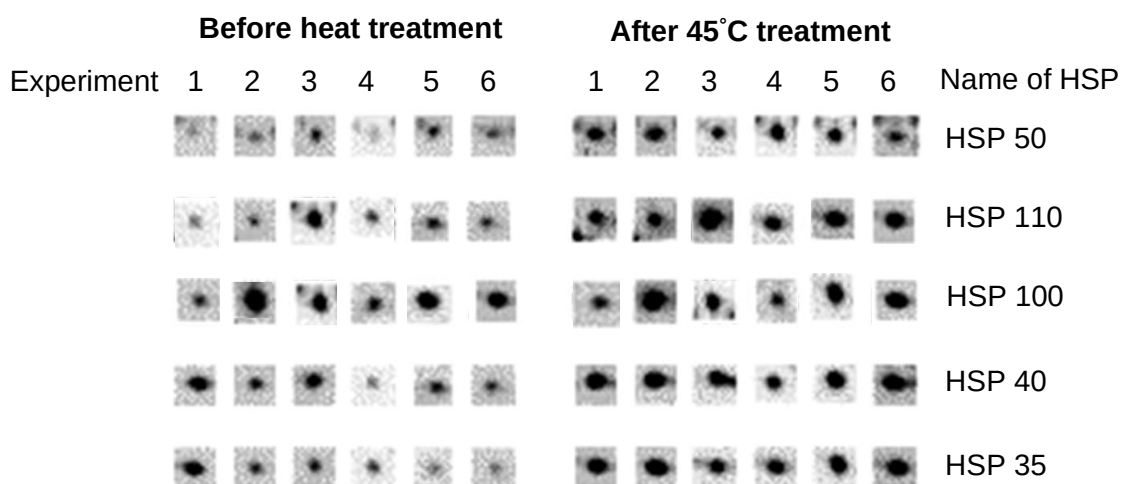


Fig. 1.4

- (ii) Using the information provided, deduce the identity of the heat shock protein that did not respond to the heat treatment.

..... [1]

- (iii) Give a reason for your answer in (c)(ii).

..... [1]

[Turn over]

Using 2D gel electrophoresis, the expression levels of different genes of an *E. coli* strain at different temperatures were determined and shown in Fig. 1.5.

Each data point represents an individual gene.

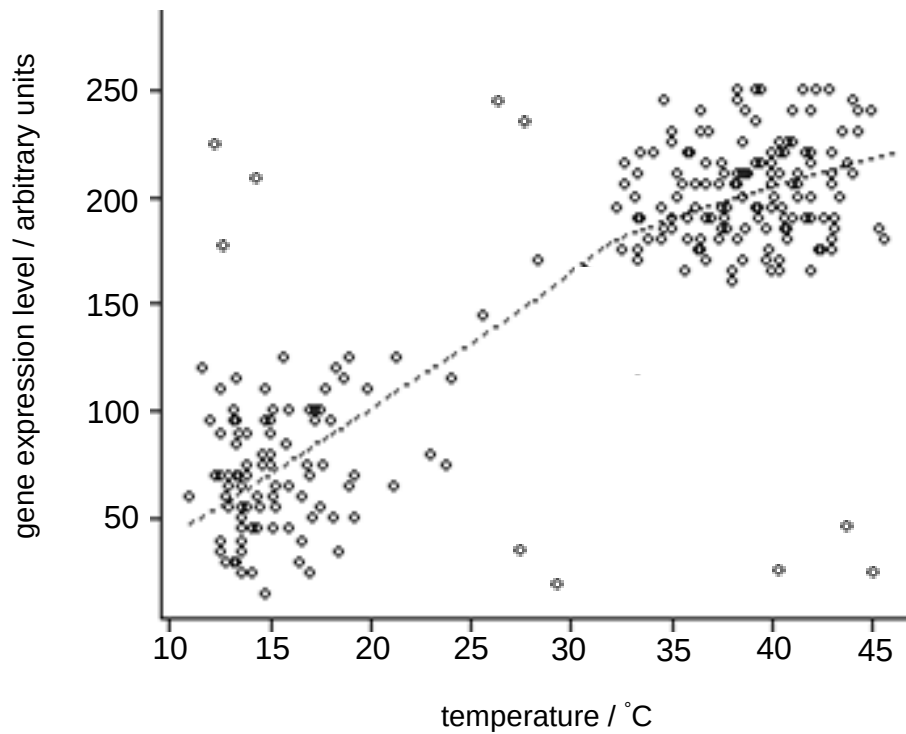


Fig. 1.5

- (iv) With reference to Fig. 1.5, describe the extent to which temperature affects the expression levels of the genes.

.....

.....

.....

.....

.....

.....

..... [3]

HSP 35 is known to be a protease. It specifically hydrolyses the peptide bond between phenylalanine (Phe) and alanine (Ala). Thus, a small tripeptide consisting of glycine, phenylalanine and alanine, Gly-Phe-Ala, is hydrolysed to Gly-Phe and Ala.

- (v) In the space below, draw a labelled diagram to illustrate the above enzymatic reaction. Include the respective R groups of the amino acids in your answer.

Legend:

R group of Gly: $-H$

R group of Ala: $-CH_3$

R group of Phe: $-CH_2$



[3]

L

[Turn over]

- (d) Scientists produced genetically engineered bacterial cells. The gene coding for the green fluorescent protein (GFP) is joined upstream of the heat-shock gene.

Promoter	Operator	GFP gene	Heat-shock gene
----------	----------	----------	-----------------

Fig. 1.6

Both genes are transcribed to produce a single mRNA molecule. When translated, a “fused” protein is produced.

- (i) Describe one feature of the control of eukaryotic transcription that is **not** shown in Fig. 1.6.

.....

.....

.....

.....

..... [2]

The “fused” protein shows up as a green fluorescent spot when viewed using a fluorescent microscope. The amount of the “fused” protein may be determined by the intensity of the green fluorescence.

- (ii) Suggest one advantage of this method over 2D gel electrophoresis in determining the amount of protein.

.....

.....

..... [1]

- (e) Besides having operons, bacteria have other means to adapt to changing environments through horizontal gene transfer. A method is through bacterial transformation.

- (i) Outline the process of bacterial transformation.

.....

.....

.....

.....

..... [2]

Insulin injection is required by diabetic patients on a daily basis. To produce sufficient insulin to meet the global demand, scientists inserted the human insulin gene into a bacterial plasmid DNA to form a recombinant plasmid. This gene was derived by reverse transcribing the mature mRNA.

The recombinant plasmid is introduced into bacterial cells by transformation that involves a chemical method. Upon expression of the human insulin gene, the insulin protein is then extracted from the bacterial cells and purified.

The production of human insulin using bacteria is shown in Fig. 1.7.

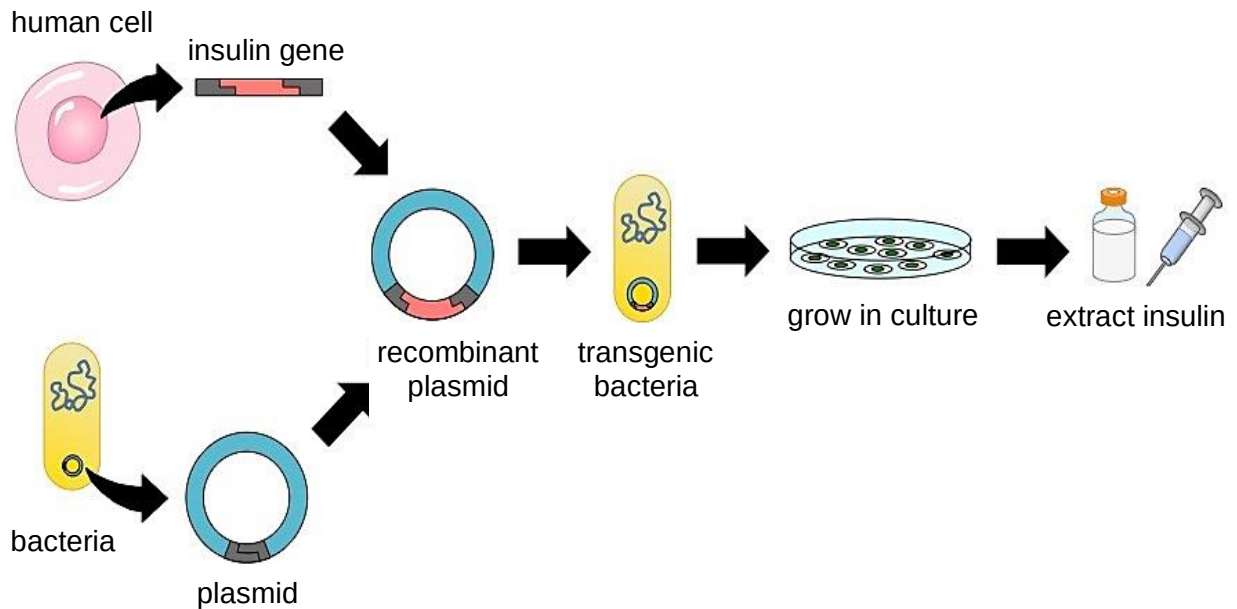


Fig. 1.7

(ii) Suggest why bacteria is used for the production of human insulin.

.....

.....

..... [1]

A young scientist wanted to synthesise human insulin in his own laboratory. He extracted the human insulin gene successfully. Before the human insulin gene was inserted into the plasmid, he removed its promoter.

He then inserted the gene at a site in a plasmid that is just downstream of a *lac* operon promoter which he had also inserted into the plasmid.

- (iii) Explain why he decided to use *lac* operon promoter instead of relying on the promoter of the human insulin gene.

.....
.....
..... [1]

- (iv) The young scientist followed the steps shown in Fig. 1.7 carefully. However, the gene used was extracted directly from nuclear DNA.

Suggest why the scientist is unlikely to obtain functional insulin even if the gene was transcribed successfully.

.....
.....
.....
.....
..... [2]

[Total: 25]

- 2 The Hawaiian Islands are ordered by age, with the oldest islands in the North-West and the youngest in the South-East.

Fig. 2.1 shows the main Hawaiian Islands together with their date of formation, millions of years (My) ago.

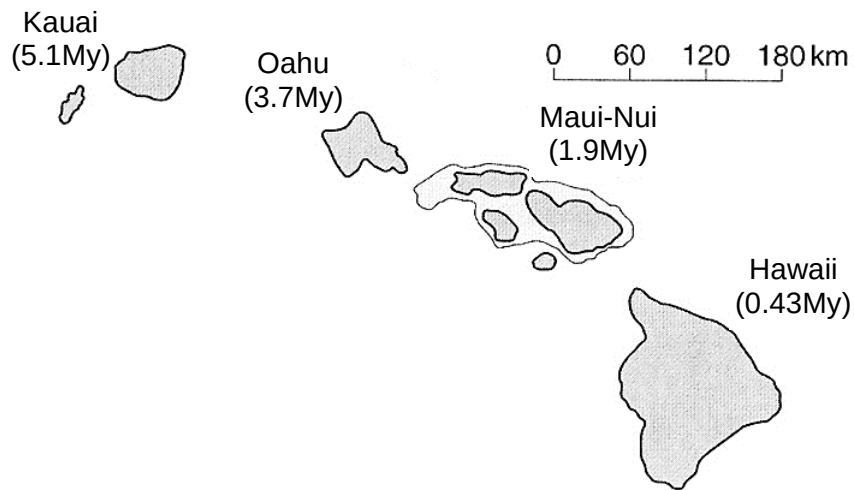


Fig. 2.1

Three closely related species of honeycreeper (a type of bird) are found on three different islands.

- Oahu Creeper, *Hemignathus kauaiensis*
- Maui Creeper, *Hemignathus flavus*
- Hawaii Creeper, *Hemignathus virens*

The date of an island's formation provides a maximum age for species inhabiting the island, assuming that populations colonise soon after the island's emergence.

1

[6]

- (b) Differences in the DNA sequences that encode cytochrome b within the individual species have been measured. Cytochrome b is a protein found in the electron transport chain.

Fig. 2.2 shows differences in cytochrome b DNA within the individual species plotted against time since colonisation (My).

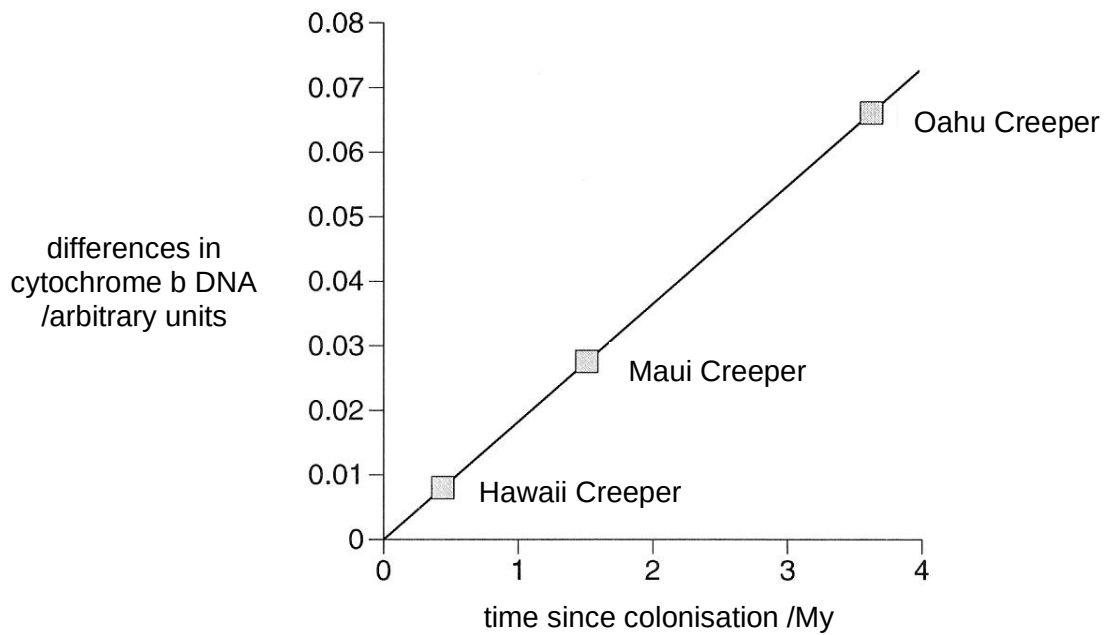


Fig. 2.2

- (iv) Suggest why the cytochrome b gene is used to measure changes in DNA sequences in closely related species.

.....

 [1]

- (v) Explain the trend in the differences in cytochrome b DNA within the individual species.

.....

 [2]

- (c) Fossils provide strong evidence for Darwin's theory of evolution.

A fossil named *Ambulocetus* was discovered in 1992 in Pakistan, and it was dated at roughly 47 – 48 million years old. Paleontologists (scientists who study the history of life on Earth through fossil records) believe that this fossil is a transitional fossil between ancestral land mammals and present-day cetaceans (aquatic mammals, e.g. dolphins, whales).

Fig. 2.3 shows an artist's impression of *Ambulocetus*.



Fig. 2.3

- (i) With reference to Fig. 2.3, describe two structural features of the *Ambulocetus* that would support the hypothesis that it is a transitional fossil.

.....

.....

.....

.....

..... [2]

- (ii) Explain why transitional fossils provide strong evidence for Darwin's theory of evolution.

.....

.....

.....

.....

..... [2]

[Total: 13]

- 3 Fig. 3.1 shows the seaweed *Laminaria hyperborea*. This is a photosynthetic organism found in the coastal waters around Norway.

The seaweed is grown commercially to obtain the polysaccharide, alginate. Alginate is made up of repeating units of glucose. This is used in certain food products.



Fig. 3.1

An increase in carbon dioxide concentration in the atmosphere has resulted in higher concentrations of carbon dioxide in the ocean. This has caused a decrease in the pH of the ocean and has resulted in ocean acidification.

Scientists are studying seaweeds such as *L. hyperborea* because they absorb a large quantity of carbon dioxide during photosynthesis. This may help to increase the pH of the ocean and reverse ocean acidification.

- (a) Outline the reactions occurring in the stroma of the chloroplast that lead to the production of a polysaccharide, such as alginate.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Laboratory experiments were carried out to investigate the effect of day length on the rate of photosynthesis in another marine organism, *Zostera marina*.

- The temperature was controlled at 4 °C.
- A low concentration of carbon dioxide dissolved in the water was used.
- The light exposure period (day length) was different for five groups of *Z. marina*.
- This was maintained for 10 days to allow *Z. marina* to adapt to these conditions.
- After 10 days, the rate of photosynthesis was measured for each group under the **same** controlled conditions.
- The experiment was repeated using five groups of *Z. marina* with a high concentration of carbon dioxide dissolved in water.

Table 3.1 shows the rate of photosynthesis for each group.

Table 3.1

day length / hours	rate of photosynthesis / arbitrary units	
	low carbon dioxide concentration	high carbon dioxide concentration
12	2.0	2.5
14	3.0	5.0
16	4.0	7.0
18	5.5	11.0
20	7.5	18.0

(i) With reference to Table 3.1, explain the difference in the rate of photosynthesis at high carbon dioxide concentration compared to low carbon dioxide concentration.

.....

 [1]

(ii) With reference to Table 3.1, describe **and** explain the effect of increasing day length on the rate of photosynthesis for the *Z. marina* in high carbon dioxide concentration.

.....

 [4]

- (c) In the laboratory, a seaweed was grown in water with different pH values. All other variables, including temperature and light, were standardised.

The mean rate of photosynthesis was calculated over a 24 hour period for each pH value.

The results are shown in Fig. 3.2.

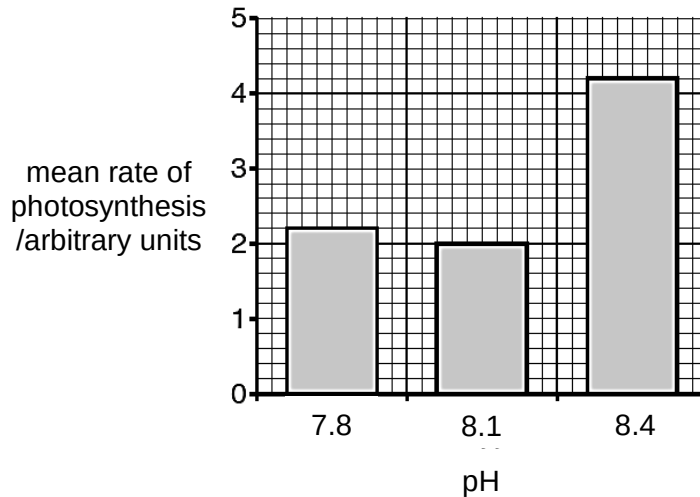


Fig. 3.2

- (i) With reference to Fig. 3.2, explain the effect on the rate of photosynthesis when the pH increases from 8.1 to 8.4.

.....

.....

.....

.....

..... [2]

- (ii) The lower pH values on Fig. 3.2 represent ocean acidification.

Suggest why the results for the lower pH values do **not** fully support the idea that seaweeds can help to reduce ocean acidification.

.....

.....

..... [1]

[Total: 12]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)** as indicated in the question.

- 4 (a) Outline the structures and functions of named organelles involved in the synthesis of polypeptides in eukaryotes. [10]

- (b) Smallpox is an infectious disease that was eradicated in the 20th century with the aid of a successful global vaccination effort. However, there remains many other infectious diseases that could not be eradicated yet, and the human immune system is always in a constant race to produce a diversity of antibodies against these known and unknown pathogens.

Discuss the factors that led to a successful eradication effort against smallpox disease and describe how the immune system is able to rapidly and strongly respond to a wide variety of pathogens through the production of antibodies.

[15]

[Total: 25]

- 5 (a) Explain how fluidity of biological membranes can be maintained and the importance of membrane fluidity in biological processes. [10]

- (b) Genetic diseases, such as sickle cell anaemia, are caused by gene mutations. Although many of these diseases can be identified through their symptoms, doctors tend to perform further tests in order to accurately diagnose them. These additional tests involve the use of molecular techniques.

Explain how gene mutations can result in sickle cell anaemia, and account for how molecular techniques can be used for the detection of this disease.

[15]

[Total: 25]







This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

BLANK PAGE

